GHRaisoni International Skill Tech University, Pune

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

1. Use Case Diagram

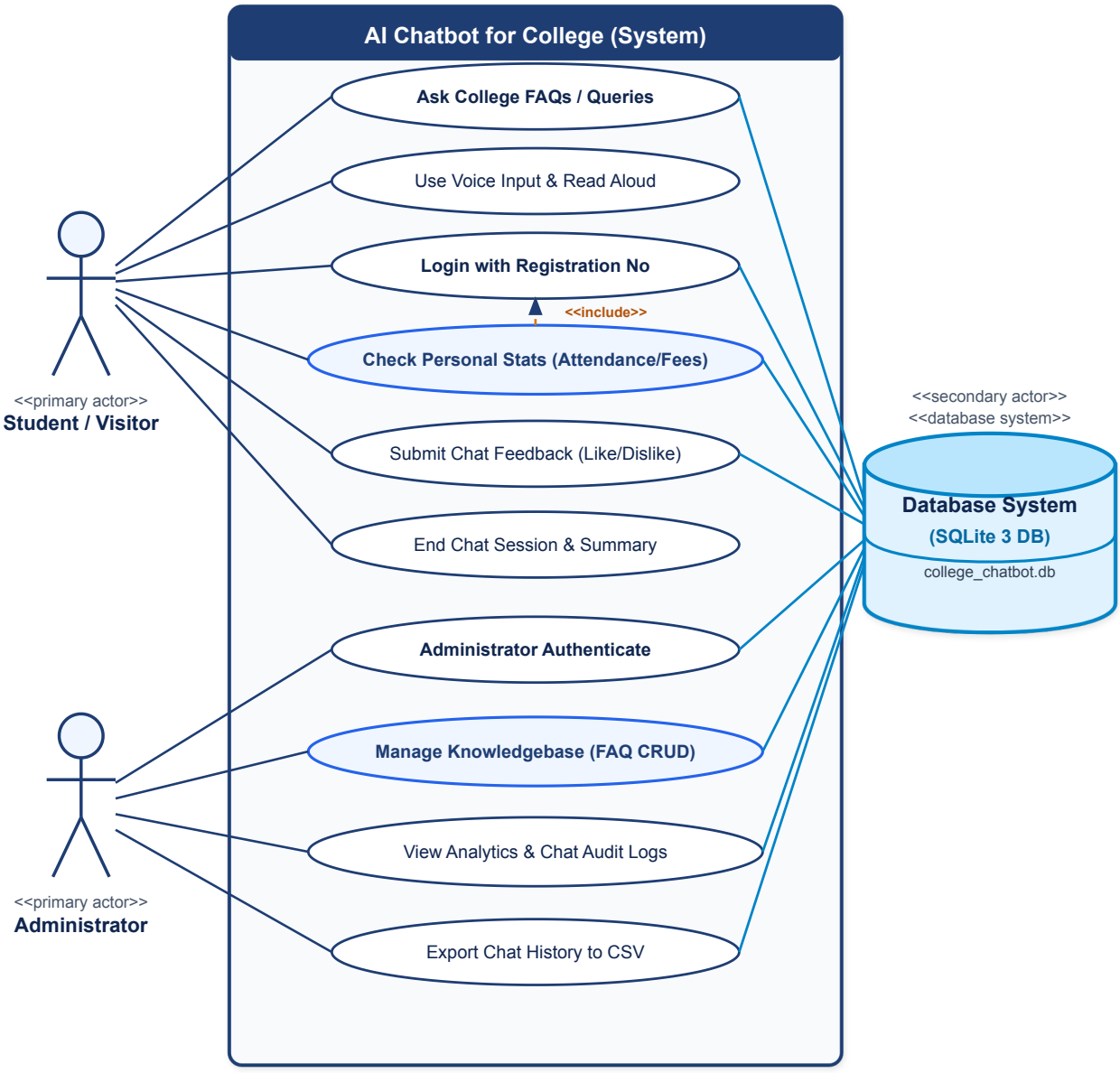


Fig 1: Use Case Diagram — primary actors (Student & Administrator), system boundary, and broad secondary database cylinder (SQLite 3 DB) with data-tier associations.

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

2. Class Diagram (With Full Relations & Multiplicities)

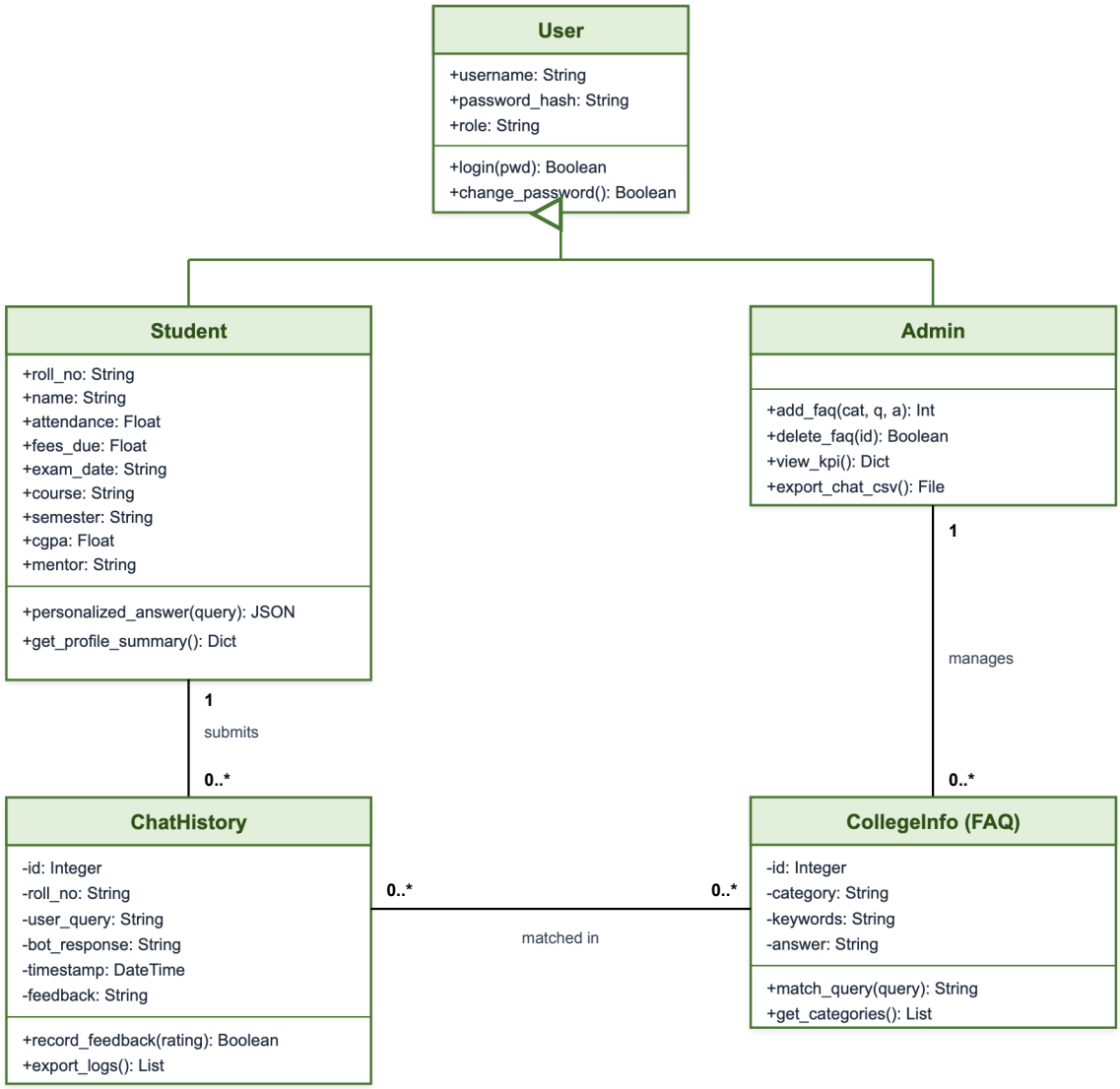
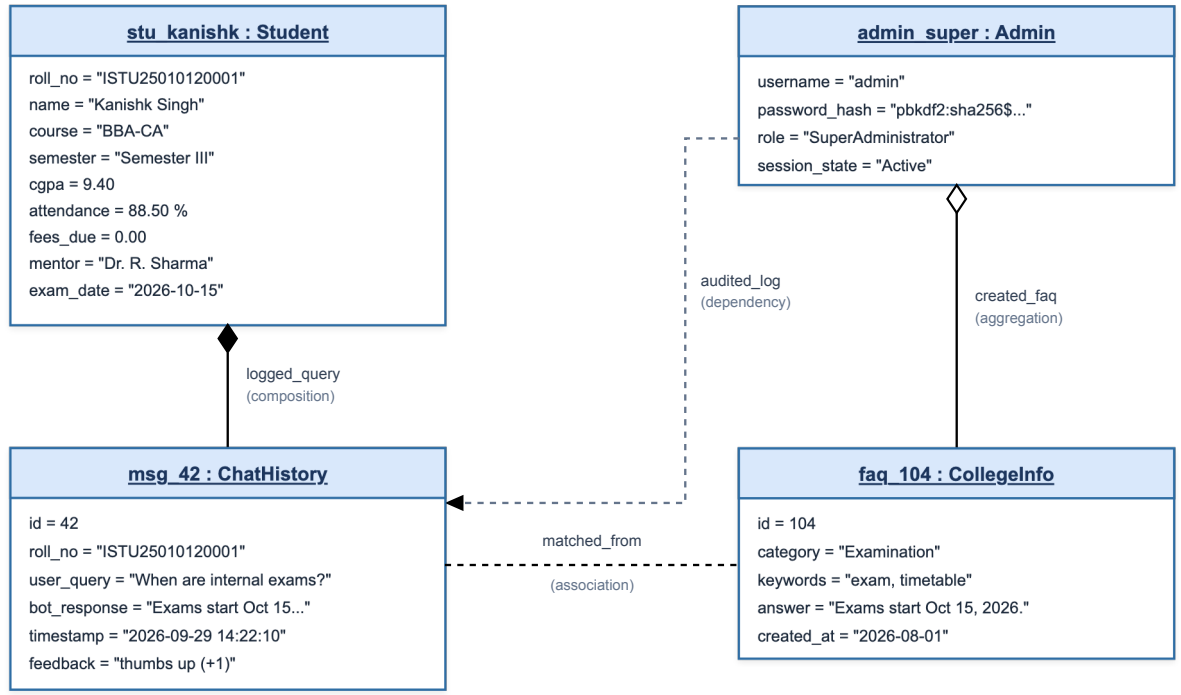


Fig 2: Class Diagram — User generalization with Student and Admin subclasses, association to ChatHistory (submits) and CollegeInfo FAQ (manages), and search association (matched in).

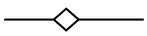
SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

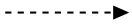
3. Object Diagram (Runtime Instances & Slot Values)



Object diagram notations and their uses



Aggregation
Part can exist without whole
Admin to CollegeInfo



Dependency
One object uses another
Admin to ChatHistory



Composition
Part dies with the whole
Student to ChatHistory

Link
Plain line joining two objects



Object box
Instance : class with slot values

Association
Directed link, ChatHistory to FAQ

Fig 3: Object Diagram — runtime instances (stu_kanishk, admin_super, msg_42, faq_104) with composition, aggregation, dependency, association, and formal notation guide.

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

4. Sequence Diagram — Query, Feedback & Session Termination

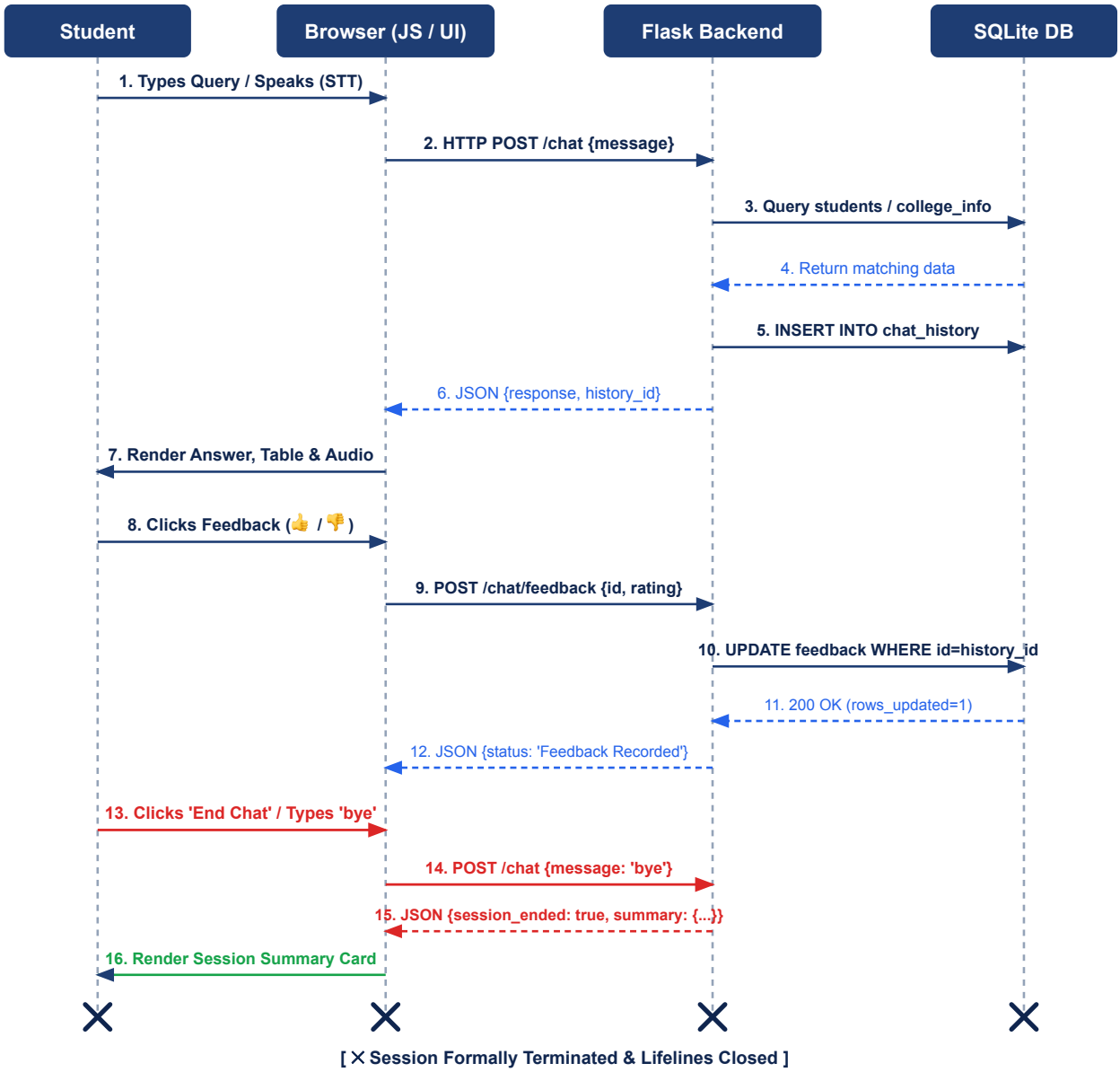


Fig 4: Sequence Diagram — complete query, feedback ACK, explicit End Chat flow, and standard UML lifeline termination (X).

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

5. Activity Diagram — Chatbot Response & Session Flow

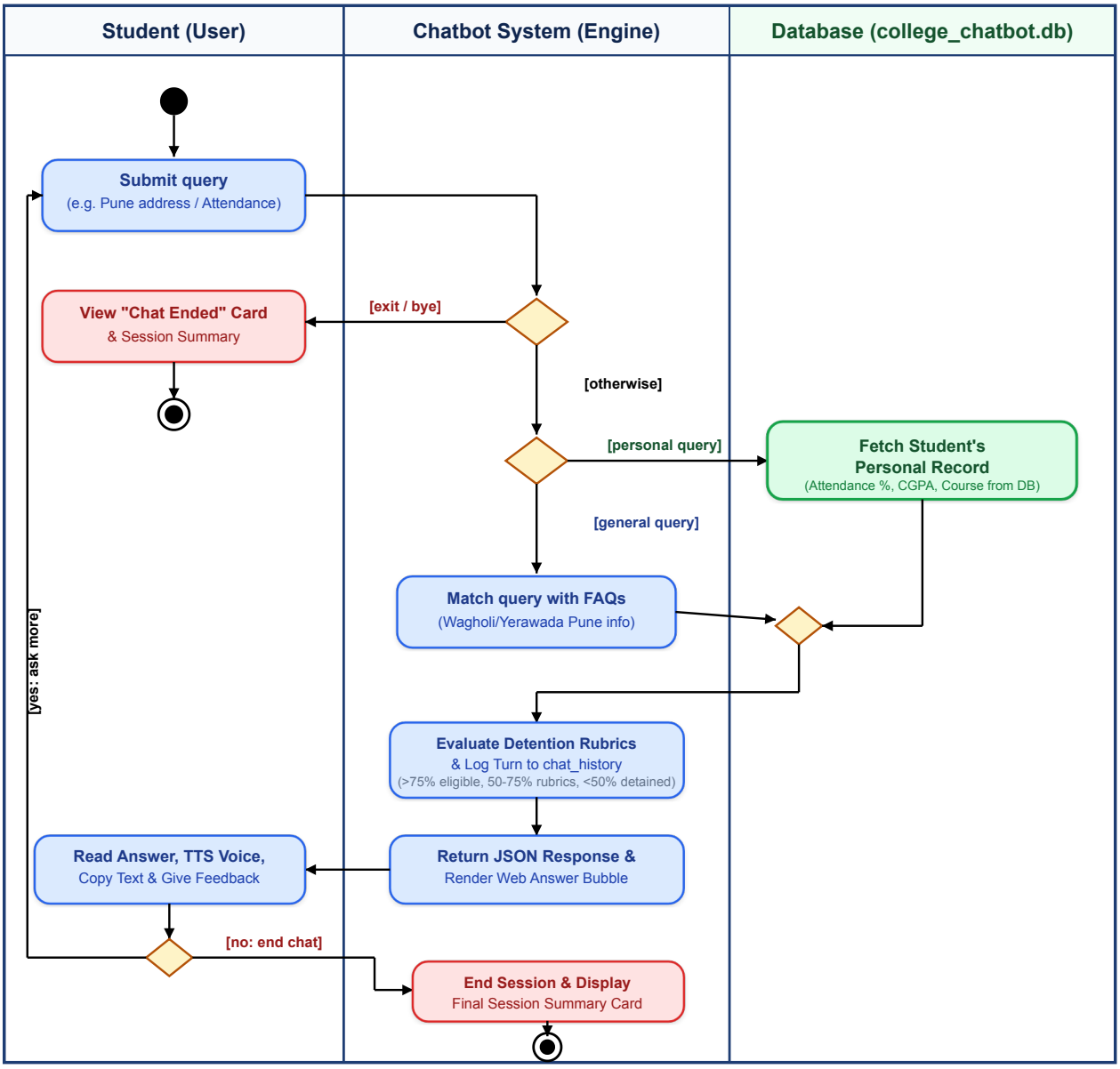
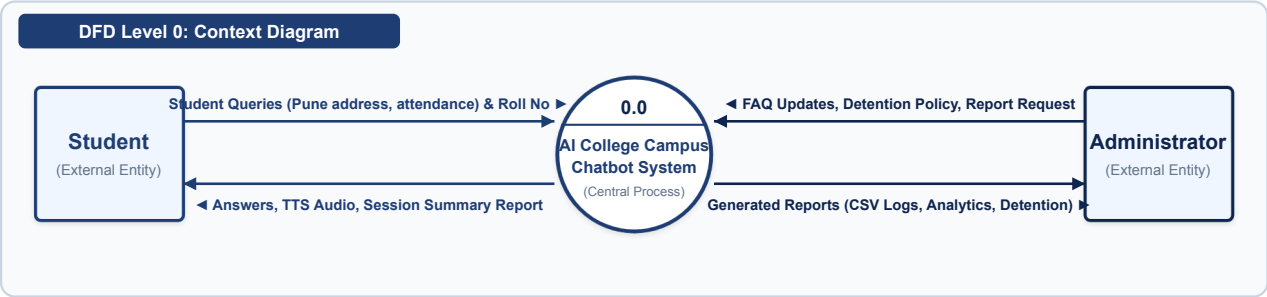


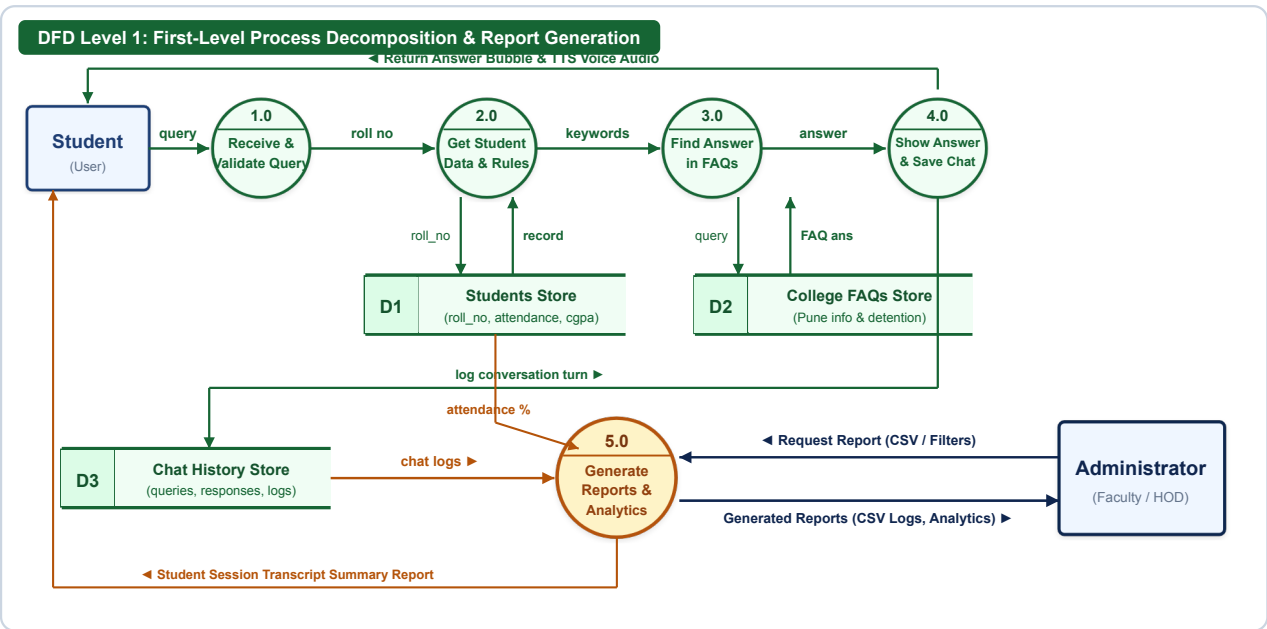
Fig 5: Activity Diagram — 3-Swimlane architecture (Student, Chatbot System, Database) explicitly showing database fetching of student personal records, FAQ matching, and controlled session loop with explicit End states.

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

6. Data Flow Diagram (DFD Level 0 Context & Level 1 Process Breakdown with Report Generation)





Standard DFD Notation & Symbol Conventions (Gane-Sarson & DeMarco)

Symbol Shape	DFD Element	Standard Academic Definition	Implementation in Our Project
▭	**External Entity**	Origin/destination of data outside boundary (Square)	**Student (User) & Administrator (Faculty/HOD)**
○	**Process (Bubble)**	Transforms input data into output data (Circle with ID) Calculates, retrieves, formats, or generates reports	**1.0 to 4.0 Query Handlers & 5.0 Generate Reports** Query answering & administrative CSV/analytics reporting
D ▭	**Data Store** *(Open-Ended)*	Repository of data at rest (Open parallel lines) **MANDATORY RULE: Right side must be open; NEVER closed**	**D1 Students, D2 College FAQs, D3 Chat History** Persistent SQLite tables feeding search & report generation
→	**Data Flow**	Directed solid arrow showing data path & payload Transfers packets between entities, processes, and stores	**query, record, FAQ answer, generated reports CSV** Directional vectors with explicit noun-phrase labels

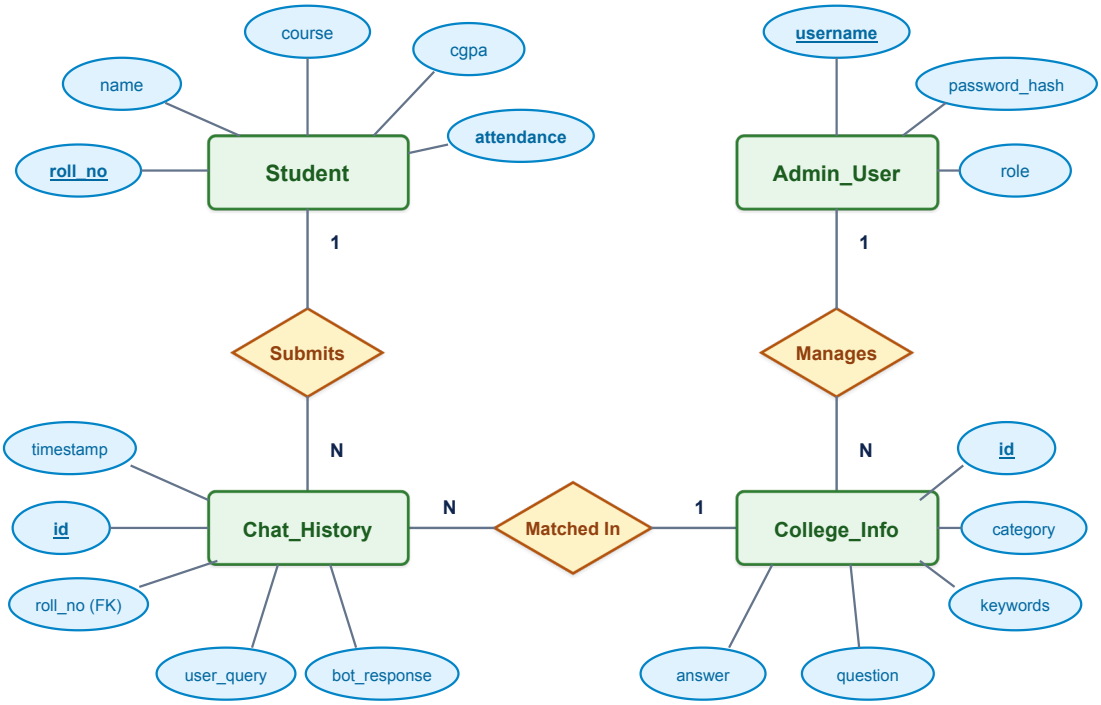
Fig 6: Data Flow Diagram (DFD) — Properly formatted Level 0 Context Diagram (System boundary with Student & Administrator) plus Level 1 Process Decomposition (1.0 to 4.0 Query Handlers & 5.0 Generate Reports & Analytics) with standard open-ended data stores (D1 Students, D2 FAQs, D3 Chat History) and formal academic DFD symbols reference table.

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

7. Entity-Relationship Diagram (ERD — Database Architecture)

ER Diagram (Classic Notation): AI Chatbot for College



Schema & Project Knowledge Base Defense:

- **Student Entity:** Stores attendance %, CGPA & evaluates 4-tier detention policy.
- **College_Info Entity:** Contains Pune campuses (Wagholi & Yerawada) & exam rules.
- **Chat_History Entity:** Logs each turn with student roll_no FK, query & bot answer.
- **Admin_User Entity:** RBAC authentication for updating college FAQs & detention data.

Verified in 3NF Normalization: Zero transitive dependencies & lossless join preservation.

Key / Legend:

- Entity (Table)
- Attribute (Column) — id = Primary Key
- Relationship
- Link / Cardinality (1:N)

Fig 7: Entity-Relationship Diagram (ERD — Classic Notation) — 4 core entities (Student, Admin_User, Chat_History, College_Info) with attributes, cardinalities (1:N, N:1), and official symbol key.

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

8. Deployment Diagram (Hosting Topology & Artifact Architecture)

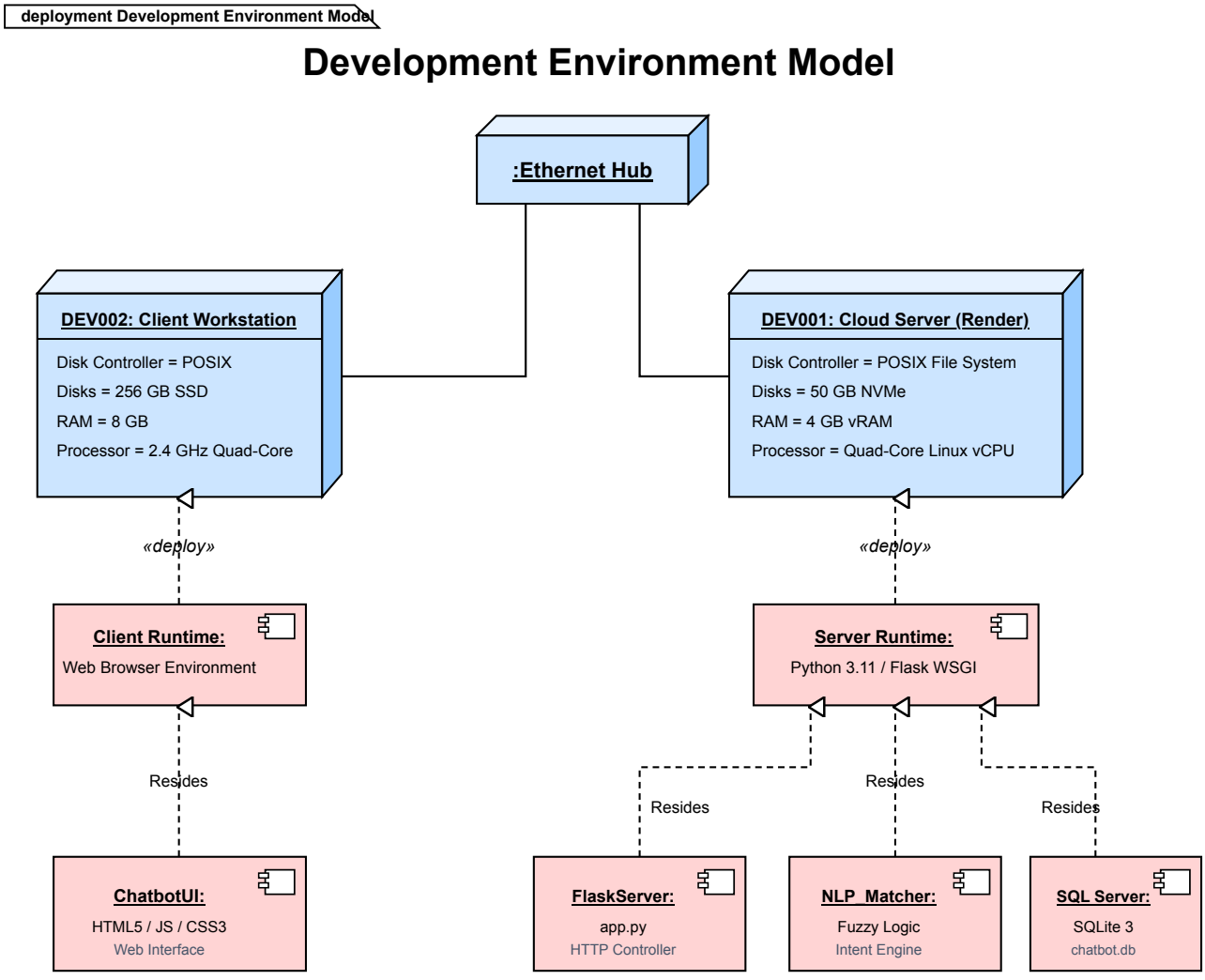


Fig 8: Deployment Diagram — Development Environment Model with :Ethernet Hub, 3D hardware nodes with specifications, «deploy» dependency arrows, and resident software component boxes.